LITTLE STREAM DETECTOR

LSD 3.0

Detailed Technical Documentation

Property	Value
Document type	Architecture and implementation reference
Release	3.0 - VP9 validated build
Reference file	LSD-3.0-VP9-title-final.ps1
Build SHA-256	a6dc7cc2af9ece7c3e219a72aefee97822f8042a0d4c31dfda3982664f8a5590
Documentation date	2 September 2026
Runtime	Windows PowerShell 5.1 and .NET Framework
External multimedia tools	None
Temporary media files	None

Scope. This document describes the native container, codec, canonical-sample, analysis, validation, reporting, threading, safety, and dual-result comparison architecture of LSD 3.0, including the validated native VP9 frame and Base Q Index parser.

Release principle. Version 3.0 keeps one canonical parser path. The VP9 extension is isolated within codec analysis and consumes the existing bounded canonical sample model. Current A and Reference B continue to use immutable completed-result snapshots.

Contents

Updated in this revision. VP9 is now a supported native video family for Matroska/WebM. Documentation now covers VP9 superframes, hidden coded frames, show-existing events, KEY/INTER classification, reference-slot geometry, Base Q Index extraction, completeness checks, and the validated regression assets.

1. Purpose and System Scope

Little Stream Detector 3.0 is a native Windows PowerShell application with an embedded C# analysis engine. It inspects media containers, builds a container-neutral sample index, performs codec-aware bitstream analysis, and presents human-readable and structured results through a responsive graphical interface.

The VP9 extension adds deterministic traversal of VP9 uncompressed frame headers without changing container addressing, the canonical model, or A/B comparison semantics.

1.1 Primary objectives

- Provide deterministic native analysis without FFmpeg, MediaInfo, or other external multimedia processes.
- Keep container-specific addressing separate from codec-specific interpretation.
- Validate every canonical sample against file boundaries before reading payload data.
- Report unavailable information as N/A rather than guessing.
- Support asynchronous preparation and analysis without blocking the GUI.
- Retain Windows PowerShell 5.1 and in-memory EXE hosting compatibility.
- Allow visual comparison of exactly one Current A and one Reference B using completed native results.
- For VP9, distinguish container samples, internal coded frames, shown frame events, and hidden frames.

1.2 Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, VP9, MPEG-4 Part 2/XviD
Audio	AAC-LC, HE-AAC, HE-AACv2, MP3, PCM integer/float, AC-3, E-AC-3
Outputs	Summary A, optional Summary B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Architectural Overview

LSD remains a modular monolith. PowerShell owns application state, presentation, orchestration, and background-job lifecycle. Embedded C# types own binary parsing, canonical models, bitstream traversal, validation, and numerical accumulation.

Pipeline. User/GUI -> metadata and job coordinator -> Matroska, MP4, or AVI reader -> canonical media model -> video and audio analyzers -> completed result snapshot -> Current A / Reference B comparison views.

2.1 Logical layers

Layer	Responsibility	Representative components
Presentation	Window, reports, graphs, drag-and-drop, active-tab copy	PowerShell WinForms
Coordination	Metadata preparation, state transitions, cancellation, progress	Start-Meta, Complete-NativeMetadata, Start-Scan
Container parsing	Tracks and canonical sample locations	NativeMediaProbe, NativeMp4Probe, NativeAviProbe
Canonical adaptation	Normalize tracks and bounded samples	MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter
Codec analysis	Configuration and frame-level syntax	AVC, HEVC, AV1, VP9, MPEG-4 Visual
Comparison	Retain B and render A/B views	ReferenceResult, ReferenceOnlyView

2.2 VP9 isolation

The V_VP9 mapping already produces CodecFamily VP9 and PayloadFormat Vp9Frame. The new Vp9FrameParser is selected only when the active video track is VP9. Other codec branches and all container readers remain unchanged.

3. Runtime and Processing Lifecycle

3.1 Startup

- Compile embedded C# through Add-Type.
- Run native self-tests for selected readers and arithmetic behavior.
- Hide the console window for GUI operation.
- Apply invariant English culture for deterministic formatting.
- Initialize current-result, reference-result, VP9 diagnostics, jobs, graph, and report state.

3.2 File processing phases

Phase	Action	Output
1. Selection	Select or drop a file	Absolute source path
2. Native detection	Identify container	NativeMediaInfo
3. Metadata preparation	Parse tracks and sample indexes	Container probe result
4. Canonical adaptation	Create common bounded samples	LsdMediaModel
5. Codec analysis	Parse frame and audio syntax	Diagnostics and statistics
6. Finalization	Build report and histograms	Current A result
7. Optional promotion	Set Ref B	Reference B snapshot
8. Comparison	Analyze later A	A/B views

3.3 VP9 sample lifecycle

Each bounded Vp9Frame sample is read once. A valid superframe index may split that sample into multiple internal VP9 frames. Each internal frame is parsed in coded order so reference-slot geometry remains available. The sample contributes one bitrate observation, while every newly coded internal frame contributes one Base Q Index observation.

4. Canonical Media Model

4.1 LsdMediaModel

Field	Meaning
SourcePath	Input file path
ContainerType / ContainerDetail	Normalized family and specific route
FileLength	Source length in bytes
DurationSeconds	Best available duration
TimecodeScale	Nanoseconds per canonical timestamp tick
Tracks	Canonical video, audio, subtitle, and other tracks

4.2 LsdTrackModel and LsdSampleModel

Object	Key fields
LsdTrackModel	TrackId, TrackType, CodecId, CodecFamily, CodecPrivate, PayloadFormat, NalLengthSize, PCM parameters, Samples
LsdSampleModel	FileOffset, Length, DecodeTimestamp, PresentationTimestamp, Duration, key/invisible/discardable flags, PayloadFormat

4.3 VP9 accounting semantics

A canonical VP9 sample is a container payload range, not necessarily one coded frame. Superframes can carry one shown frame and one or more hidden coded frames. LSD therefore reports Video samples, Internal VP9 frames, Shown frame events, Hidden coded frames, and Show existing frame separately.

4.4 Reference result snapshot

Reference B remains a presentation-layer snapshot containing file identity, duration, bitrate bins, quantizer mode, quantizer counts, and frozen Summary B text. VP9 uses QpMode "VP9 Base Q Index".

5. Container Modules

5.1 Matroska and WebM

The Matroska reader parses EBML elements, Segment metadata, Tracks, Cluster timing, Block and SimpleBlock structures, and supported lacing modes. V_VP9 is normalized to CodecFamily VP9 and PayloadFormat Vp9Frame. Canonical sample payload ranges are verified before frame traversal.

5.2 MP4, M4V, and MOV

The ISO Base Media reader supports non-fragmented files with a moov-based sample table. It derives sample offsets, sizes, decode times, composition offsets, and sync flags from stsd, stsz, stsc, stco/co64, stts, ctts, and stss. VP9 sample-entry mapping is not enabled in this build.

5.3 AVI 1.0 and OpenDML AVI 2.0

The AVI module reads RIFF headers, stream formats, LIST/INFO metadata, movi areas, legacy idx1 entries, AVIX segments, and standard ix## indexes. VP9 in AVI is not part of the validated support matrix.

5.4 Comparison neutrality

Reference comparison never reads container structures. Completed VP9 histogram data are compared only through the same normalized result snapshot used by other codecs.

6. Video Analysis Modules

Family	Native analysis
H.264 / AVC	avcC and Annex B, SPS/PPS, I/P/B, SliceQPY, CABAC/CAVLC, VUI
H.265 / HEVC	hvcC, VPS/SPS/PPS, NAL class, first-slice I/P/B, SliceQPY, VUI
AV1	av1C, sequence/frame OBU traversal, hidden/show-existing, Base Q Index
VP9	Superframe index, uncompressed header, reference slots, shown/hidden events, KEY/INTER, Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/non-coded VOPs, packed samples, VOP quantizer

6.1 VP9 parser scope

- Validate the two-bit frame marker and profile signaling.
- Recognize show_existing_frame without inventing a new quantizer value.
- Parse KEY_FRAME, INTER_FRAME, and intra-only syntax needed to reach quantization.
- Track eight reference slots so frame_size_from_refs can resolve width and height.
- Split valid superframes using the trailing duplicated marker and little-endian frame sizes.
- Read base_q_idx and the three frame-level delta-Q fields.
- Classify displayed KEY frames as I and displayed INTER frames as P. VP9 has no classic B-frame class in this report model.

6.2 Quantizer meaning

VP9 Base Q Index is a frame-level lookup-table index from 0 through 255. Segmentation may override quantization for individual blocks, so LSD does not label this histogram as a block-level quantizer distribution.

7. Audio Analysis Modules

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate and channels, SBR, Parametric Stereo, profile naming
MP3	Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples
AC-3 / E-AC-3	Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE
PCM	Block alignment, payload divisibility, decoded duration, expected and measured bitrate

Audio analysis remains independent of the VP9 extension. Audio details for Reference B remain available through Summary B. No waveform or quality score is synthesized.

Regression observation. The full VP9 reference file retained AAC-LC accounting with 27,247 canonical samples, zero rejected samples, and a measured bitrate of 128.000 kbps.

8. Quantizer and Bitrate Analysis

8.1 Bitrate profile

Each canonical video sample contributes byte size to presentation-time bins. Internal hidden VP9 frames remain inside their containing sample and do not create synthetic timing or bitrate bins. Reference B is gold and Current A is blue.

8.2 Quantizer sources

Codec	Native value	Histogram unit
AVC	SliceQPY	Frame-level SliceQPY
HEVC	SliceQPY	Frame-level SliceQPY
AV1	Base Q Index	New coded frame
VP9	Base Q Index	New internal coded frame
MPEG-4 Part 2	VOP quantizer	Coded VOP

8.3 VP9 completeness

- A Base Q Index is emitted only for a newly coded VP9 frame.
- show_existing_frame produces a display event but no new Base Q Index.
- A superframe can emit multiple Base Q Index values from one canonical sample.
- The histogram is enabled only when rejected headers equal zero and at least one internal coded frame was parsed.
- Frame accounting requires InternalFrames = Parsed + ShowExisting.
- Shown/sample consistency requires ShownFrameEvents = VideoSamples for the validated Matroska route.

8.4 Dual DRF/QP distribution

The right panel uses the current quantizer mode in its title. For VP9 it displays "VP9 Base Q Index distribution". The contextual range is clamped to the valid codec domain 0 through 255.

9. GUI, Jobs, Reporting, and Comparison

9.1 Views

View	Content
Summary A	Readable report for Current A, including VP9 diagnostics
Summary B	Frozen reference report while B exists
Streams	Structured current native-track data
JSON	Serializable current metadata and streams
Log	Current lifecycle and validation messages
Bitrate graph	Blue A and gold B with role legend
Quantizer panel	Mode-aware A/B statistics, bars, and percentages

9.2 VP9 report block

The Native VP9 Frame / Base Q Analysis block reports samples, internal frames, superframes, parsed and rejected headers, show-existing events, shown and hidden frames, KEY/INTER counts, Base Q statistics, first failure, and four explicit validation statuses.

9.3 Copy action

Copy follows the active report tab. VP9 adds no alternate copy path and does not alter Summary B freeze behavior.

10. Reference Comparison State Model

The workflow supports exactly one Current A and one Reference B. It is not a queue and does not rank files.

State sequence. EMPTY -> open -> A blue -> Set Ref B -> B gold -> open -> B gold + A blue -> Remove Ref B -> A blue, or EMPTY when B was the only result.

10.1 State rules

- Every newly opened or dropped file becomes Current A and is blue.
- Set Ref B promotes completed A to B, changes the role color to gold, and waits for a new A.
- Opening another file while B exists preserves B and creates a new blue A.
- Remove Ref B removes only B.
- B is never silently converted back into A.

10.2 VP9 compatibility

Two VP9 results are directly comparable when both snapshots carry QpMode "VP9 Base Q Index". AV1 Base Q Index and VP9 Base Q Index remain distinct modes even though both use a 0 through 255 index domain. LSD does not force a cross-codec overlay.

11. Validation and Defensive Parsing

11.1 Boundary checks

- Canonical samples require non-negative offsets and positive lengths.
- Offset plus length must remain within the source file.
- VP9 sample lengths must fit the in-memory parser input contract.
- A superframe index must fit inside the sample, repeat the trailing marker, contain positive frame sizes, and account exactly for the payload before the index.
- Every internal frame reader is bounded to its indexed frame length.
- Frame marker, reserved profile bits, frame sync code, RGB profile constraints, reference-slot availability, and Base Q range are validated.

11.2 Error model

A VP9 syntax failure increments Headers rejected, preserves the first failure text, marks the sample frame type as Other, and prevents histogram activation. No guessed quantizer value is substituted.

11.3 State ordering

Internal frames are parsed in coded order. Reference slots are updated only from successfully parsed newly coded frames using refresh_frame_flags. This is required for frame_size_from_refs and for show-existing classification when a referenced slot has known state.

11.4 Runtime isolation

- No external multimedia executable is launched.
- No temporary media file is created.
- The source is opened read-only with shared read access.
- The single-file script remains compatible with in-memory EXE hosting.

12. Supported Combinations

12.1 Container and video matrix

Video	Matroska/WebM	MP4/M4V/MOV	AVI
H.264 / AVC	Yes	Yes	Yes, Annex B
H.265 / HEVC	Yes	Yes	No
AV1	Yes	Yes	No
VP9	Yes, validated	No	No
MPEG-4 Part 2 / XviD	Yes	Yes	Yes

12.2 Container and audio matrix

Audio	Matroska/WebM	MP4/M4V/MOV	AVI
AAC-LC / HE / HEv2	Yes	Yes	No
MP3	Yes	Yes	Yes
PCM integer / float	Yes	Supported entries	Yes
AC-3	Yes	Yes	Yes
E-AC-3	Yes	Yes	No

12.3 Comparison combinations

Bitrate comparison can display any two completed video results. Quantizer comparison requires exactly matching QpMode. Differences in codec family, dimensions, frame rate, duration, or frame count remain visible in the summaries.

13. Known Limitations

- Fragmented MP4 using moof, traf, and trun is not supported.
- VP9 is validated for Matroska/WebM V_VP9 tracks. VP9 sample entries in MP4 and VP9 in AVI are not enabled.
- The VP9 histogram is frame-level Base Q Index, not a per-block quantizer histogram. Segment-level alternate quantizers are not expanded into block populations.
- The current VP9 regression set covers Profile 0, 8-bit YUV 4:2:0. Other valid VP9 profiles remain parser-supported by syntax but are not yet represented by a dedicated regression asset.
- AVI H.264 targets Annex B payloads; unusual length-prefixed AVC in AVI may not be accepted.
- AVI AAC and AVI E-AC-3 mappings are not enabled.
- Container-only color metadata is not used as a fallback when the codec bitstream lacks color signaling.
- Certain HEVC PPS syntax involving scaling-list data remains intentionally incomplete.
- Reference comparison does not decode pixels and does not calculate PSNR, SSIM, or VMAF.
- Reference comparison supports one A and one B only.
- Bitrate differences across unlike resolution, frame rate, or duration are factual but not by themselves a quality verdict.

14. Verification and Regression Matrix

14.1 General scenarios

Scenario	Expected result
Open supported file	Current A appears blue; summaries and graphs complete
Promote A to B	B remains gold; Summary B freezes
Open new file with B	New A is blue; B remains unchanged
Remove B while A exists	A remains
Remove B when only B exists	Window returns to startup state
Compatible QP modes	Dual statistics and bars appear
Different QP modes	Quantizer comparison unavailable

14.2 VP9 reference assets

Asset	Expected VP9 result
10-second stream-copy sample	300 samples; 325 internal frames; 25 superframe samples; 2 KEY; 323 INTER; Q range 31-179; zero rejected
Full Reset Windows 10 Password with USB.mkv	18,977 samples; 19,176 internal frames; 199 superframe samples; 135 KEY; 19,041 INTER; Q range 0-179; zero rejected

14.3 Required VP9 invariants

- InternalFrames = Parsed + ShowExisting.
- Parsed = KeyFrames + InterFrames for newly coded frames.
- ShownFrameEvents = VideoSamples on the validated Matroska assets.
- Histogram count = Parsed.
- Rejected = 0 before the VP9 histogram is presented as complete.
- Displayed I + P + B + Other = container video sample count.

15. Maintenance Guidelines

15.1 Safe extension sequence

- Add or update container-native codec mapping.
- Normalize codec family and payload format.
- Populate canonical tracks and bounded samples.
- Reuse or isolate a payload reader and analyzer.
- Expose parsed/rejected counts and first failure.
- Add report fields only when authoritative data exist.
- Create deterministic regression assets for new and existing paths.
- Keep comparison logic downstream of completed results.

15.2 Code that should remain stable

- Canonical model contracts and sample boundary validation.
- AVI/OpenDML chunk resolution and MP4 parent-bound checks.
- Matroska lacing accounting.
- Existing AVC, HEVC, AV1, and MPEG-4 parser completeness rules.
- Background job cancellation and disposal.
- N/A reporting policy.
- A/B role semantics and snapshot immutability.

15.3 Future candidates

- Fragmented MP4 support.
- Container-color fallback through MP4 colr/nclx and Matroska Colour.
- Optional AVI AAC/E-AC-3 mapping.
- Additional video codecs such as MPEG-2 Video.
- Additional VP9 regression assets for Profiles 1, 2, and 3.
- Optional textual compatibility summary for A versus B.

Appendix A. Key Types and Responsibilities

Type / state	Responsibility
NativeMediaInfo / NativeTrack	Container-native metadata and codec configuration
LsdMediaModel / LsdTrackModel / LsdSampleModel	Container-neutral tracks and bounded samples
NativeMediaProbe / NativeMp4Probe / NativeAviProbe	Container detection, metadata, and sample indexing
NativeMatroskaPacketScan	Shared canonical video analysis job; historical type name
Vp9BitReader	Bounded MSB-first VP9 bit access
Vp9FrameParser	Superframe split, frame header traversal, reference-slot state, Base Q extraction
Vp9FrameInfo / Vp9ChunkInfo	Per-frame syntax result and per-sample internal-frame collection
CanonicalAudioAnalyzer	MP3, Dolby, and PCM validation/accounting
H264SliceHeaderParser / HevcHvccContextProbe	AVC/HEVC syntax and quantizer extraction
Av1CodecConfigurationProbe	AV1 configuration and sequence data
ReferenceResult	Frozen B file, bitrate bins, QP mode/counts, and summary
PowerShell coordinator	Jobs, state, reports, graphs, comparison, and GUI

Appendix B. Terminology

Term	Definition
Canonical sample	Container-neutral record describing a bounded payload range and timing
Internal VP9 frame	One coded VP9 frame extracted directly from a sample or from its superframe payload
VP9 superframe	One sample containing multiple concatenated coded frames and a trailing size index
Shown frame event	A frame presentation caused by a newly coded shown frame or show_existing_frame
Hidden coded frame	A newly coded frame with show_frame equal to zero
show_existing_frame	Display an existing reference slot without coding a new frame or Base Q Index
KEY_FRAME	VP9 intra frame with key-frame semantics and full reference refresh
INTER_FRAME	VP9 inter-coded frame; hidden alternate-reference frames are included
Base Q Index	Frame-level native quantizer lookup index; 0 through 255 for AV1 and VP9
Current A	Most recently analyzed active result, rendered blue
Reference B	Stored comparison result, rendered gold
Complete	All required relevant units parsed without rejection

Appendix C. Release Identification

Property	Value
File name	LSD-3.0-VP9-title-final.ps1
Version	3.0 - VP9 validated build
SHA-256	a6dc7cc2af9ece7c3e219a72aefee97822f8042a0d4c31dfda3982664f8a5590
Runtime model	Single PowerShell file with embedded C#
Runtime dependencies	Windows PowerShell 5.1 and .NET Framework classes
External multimedia tools	None
Temporary media files	None
Validated VP9 route	Matroska/WebM V_VP9, Profile 0, 8-bit YUV 4:2:0

Release status. This documentation corresponds to the LSD 3.0 artifact identified above. The build retains the stable canonical architecture and A/B comparison layer and adds native VP9 superframe parsing, uncompressed-header traversal, KEY/INTER classification, shown/hidden accounting, reference-slot geometry, and frame-level Base Q Index distribution.

Validated full-file result: 18,977 canonical video samples, 19,176 internal VP9 coded frames, 199 superframe samples, 135 KEY_FRAME, 19,041 INTER_FRAME, Base Q Index range 0 through 179, and zero rejected headers.

The next major container candidate remains fragmented MP4 support based on moof, traf, tfhd, tfdt, and trun processing.